CLIENT	**:**	**RETRACTABLE ROOF FOR Mr. NILESH PATEL**
SITE	**:**	**AHEMDABAD**
LOCATION	**:**	**AHEMDABAD, GUJARAT**
NO.	**:**	**-**
DOCUMENT NO.	**:**	**DBR-R00**

R00	10-08-2023	Design report	R.H.B	H.A.B	H.A.B
Rev No.	**Date**	**Description**	**Prepared By**	**Checked By**	**Approved By**

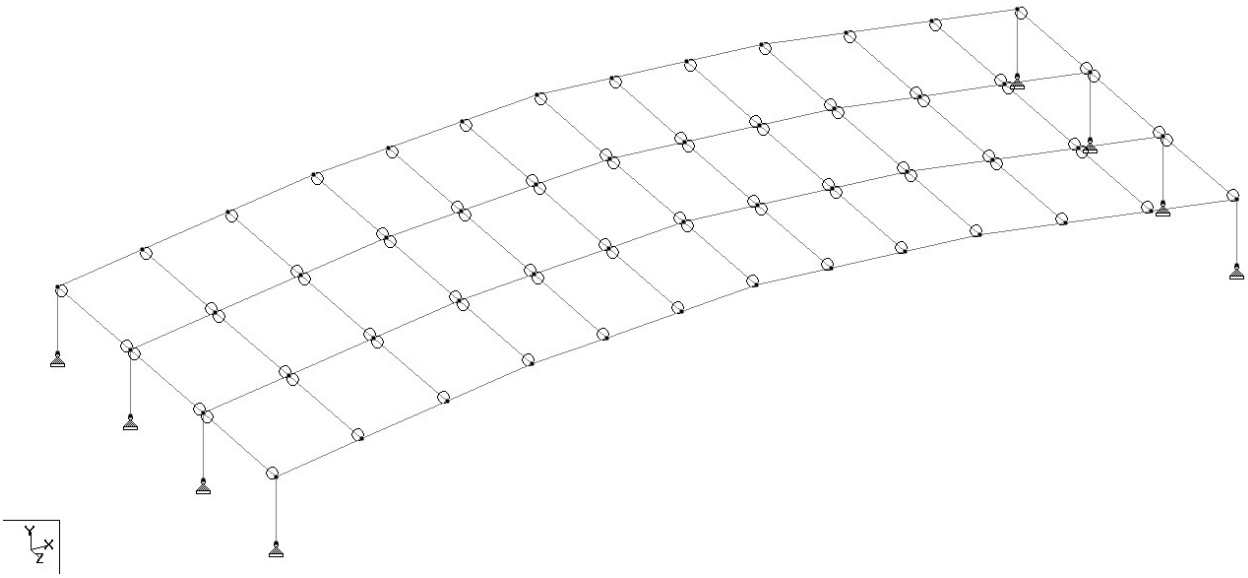
TABLE OF CONTENTS

Sr. No	Contents	Page No.
1	Design Philosophy	3
2	Design Codes and Standards	4
3	Material Specifications Construction	4
4	Load Calculations	5
4.1	Dead Load	5
4.2	Live Load	6
4.3	Wind Load Calculations	7
5	Deflection Check	8
6	Primary Loads and Load Combinations	8

1. Design Philosophy

This document pertains to structural design of **Retractable roof for Mr Nilesh patel**, Ahmedabad, Gujarat. The overall size of the frame is 9.644 x 3.81 x 1.196m high. The main sliding door frame is of steel material on which aluminium 6036-T6 material with glass is supported. The wind load is worked out from IS 875 Part III. The various DL and LL acting at different points on frame are transferred to the nearestcolumns through the grid of beams.

The complete frame is modelled as 3D frame and analysed/designed for worst combination of loads through STAAD.pro. Input and output files are enclosed which are self-explanatory. From the STAAD output, the support reactions for various primary load cases are tabulated and the base plate, shear keys and foundation bolts are designed accordingly. Foundation design shall be done using appropriate load combinations for reactions given in this document.



2. DESIGN CODES AND STANDARDS

Code / Standard	Applicability	Remark
IS800-2007	Design of steel structures	--
IS 875-Part III	For wind load calculation	--

3. MATERIAL SPECIFICATION FOR CONSTRUCTION

Sr.no	Material	Specification	Minimum Strength
1	Tubes	Indian Standard	Fy = 310 N/mm^2

4. LOAD CALCULATIONS

4.1 DEAD LOAD

SELFWEIGHT FACTOR = 1

Dead load for glass/sheet assumed = 15 Kg/m^2 i.e. 0.15 kN/m^2

4.2 LIVE LOAD

Live load assumed = 0.50 kN/m^2

4.3 WIND LOAD

For wind load calculation, following IS code is referred: - "IS 875 Part III"

DESIGN WIND PRESSURE CALCULATION				
Sr no:	**Description**	**Value**	**Unit**	**Remarks**
1	V_b Basic Wind Speed	39	m/s	As per Apendix A (IS-875 Part 3)
2	K1 (risk coefficient)	1		For Design life span of 50 yrs. As per IS 875 Part III (table-1)
3	K2 (Terrain roughness and height factor)	1		As per IS 875 Part III (table-2), Cat-2, Ht 33M
4	K3 (topography factor)	1		Clause 6.3.3 IS 875 part III
5	K4 (Importance factor for Cyclonic region.)	1		Clause 6.3.4 IS 875 part III
6	K_a (Area Averaging factor)	0.90		As per IS 875 Part III (table-4)
7	K_d (Wind Directionality Factor)	0.9		Clause 7.2.1 IS 875 part III
8	K_c (Combination Factor)	0.9		Clause 7.3.3.13 IS 875 part III
	Vz=	Vb*K1*K2*K3*K4		
	Vz	**39**	**m/s**	
	Pz =	0.91	kN/m2	Pz =0.6(Vz)^2
	Pd1 =	0.67	kN/m2	Pd1=Pz*Ka*Kd*Kc
	or Pd2 =	0.64	kN/m2	Pd2=0.7*Pz
	Hence Use pd =	0.67	kN/m2	Max of Pd1 and Pd2
	Coefficeint for Pressure Case=	1.1		
	Coefficeint for Suction Case=	0.3		
	Span =	1.2		
	UDL for Pressure case =	0.9	kN/m	
	UDL for Suction case =	0.2	kN/m	

5. Deflection check at building top in Z direction

Refer node no. –34 of staad file for 1.0 Dead load + 1.0 live load case -

Max. Deflection in Z dir. = 49.107 mm < H/180 = 9644/180 = 53.57mm (IS 800-2007 table 6)

.... Hence ok.

6. PRIMARY LOADS & LOAD COMBINATIONS

6.1 PRIMARY LOADS

Load 1 Dead Load (DL)

Load 2 Live Load (LL)

Load 3 Wind Load Pressure (WLX)

Load 4 Wind Load Suction (WLZ)

6.2 LOAD COMBINATIONS

Load Combination for strength:-

1.5 DL + 1.5 LL
1.5 DL + 1.5 WL

Load Combination for serviceability:-

1.0 DL + 1.0 LL
1.0 DL + 1.0 WL
1.0 DL + 0.8 WL

```
            **************************************************
            *                                                *
            *           STAAD.Pro CONNECT Edition            *
            *           Version  22.02.00.26                 *
            *           Proprietary Program of               *
            *           Bentley Systems, Inc.                *
            *           Date=    AUG 10, 2023                *
            *           Time=    11:31:20                    *
            *                                                *
            *  Licensed to: only @ ::LAVTeam:: (R)           *
            **************************************************


     1. STAAD SPACE
INPUT FILE: D:\SALT\1790 NIHVA MISC\PATIL\STAAD\1754 PRANAV CHANDRA-2 (15.05.2023).STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 16-DEC-22
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 14 0 7.00001 -1.05; 15 0 7.66501 -1.05; 18 4.82201 8.19501 -1.05
     9. 19 2.57401 8.05101 -1.05; 20 9.64402 7.00001 -1.05; 21 9.64402 7.66501 -1.05
    10. 22 7.07001 8.05101 -1.05; 23 0 7.00001 -4.86001; 24 0 7.66501 -4.86001
    11. 25 4.82201 8.19501 -4.86001; 26 2.57401 8.05101 -4.86001
    12. 27 9.64402 7.00001 -4.86001; 28 9.64402 7.66501 -4.86001
    13. 29 7.07001 8.05101 -4.86001; 30 0 7.00001 -2.32; 31 0 7.00001 -3.59001
    14. 32 0 7.66501 -2.32; 33 2.57401 8.05101 -2.32; 34 4.82201 8.19501 -2.32
    15. 35 9.64402 7.00001 -2.32; 36 9.64402 7.66501 -2.32; 37 7.07001 8.05101 -2.32
    16. 38 0 7.66501 -3.59001; 39 2.57401 8.05101 -3.59001
    17. 40 4.82201 8.19501 -3.59001; 41 9.64402 7.00001 -3.59001
    18. 42 9.64402 7.66501 -3.59001; 43 7.07001 8.05101 -3.59001
    19. 44 0.858002 7.79368 -1.05; 45 1.716 7.92234 -1.05; 46 3.32334 8.09901 -1.05
    20. 47 4.07267 8.14701 -1.05; 48 8.78602 7.79368 -1.05; 49 7.92802 7.92234 -1.05
    21. 50 6.32068 8.09901 -1.05; 51 5.57134 8.14701 -1.05
    22. 52 0.858002 7.79368 -4.86001; 53 1.716 7.92234 -4.86001
    23. 54 3.32334 8.09901 -4.86001; 55 4.07267 8.14701 -4.86001
    24. 56 8.78602 7.79368 -4.86001; 57 7.92802 7.92234 -4.86001
    25. 58 6.32068 8.09901 -4.86001; 59 5.57134 8.14701 -4.86001
    26. 60 0.858002 7.79368 -2.32; 61 1.716 7.92234 -2.32; 62 3.32334 8.09901 -2.32
    27. 63 4.07267 8.14701 -2.32; 64 8.78602 7.79368 -2.32; 65 7.92802 7.92234 -2.32
    28. 66 6.32068 8.09901 -2.32; 67 5.57134 8.14701 -2.32
    29. 68 0.858002 7.79368 -3.59001; 69 1.716 7.92234 -3.59001
    30. 70 3.32334 8.09901 -3.59001; 71 4.07267 8.14701 -3.59001
    31. 72 8.78602 7.79368 -3.59001; 73 7.92802 7.92234 -3.59001
    32. 74 6.32068 8.09901 -3.59001; 75 5.57134 8.14701 -3.59001
    33. MEMBER INCIDENCES
    34. 1 14 15; 3 15 44; 5 19 46; 6 20 21; 7 21 48; 8 22 50; 10 15 32; 11 18 34
    35. 12 19 33; 14 21 36; 15 22 37; 16 23 24; 17 24 52; 18 26 54; 19 27 28; 20 28 56
    36. 21 29 58; 24 32 38; 25 30 32; 26 33 39; 27 32 60; 28 34 40; 29 33 62; 31 36 42
    37. 32 35 36; 33 37 43; 34 36 64; 35 37 66; 36 38 24; 37 31 38; 38 39 26; 39 38 68
    38. 40 40 25; 41 39 70; 43 42 28; 44 41 42; 45 43 29; 46 42 72; 47 43 74; 48 44 45
```

```
39. 49 45 19; 50 46 47; 51 47 18; 52 48 49; 53 49 22; 54 50 51; 55 51 18; 56 52 53
40. 57 53 26; 58 54 55; 59 55 25; 60 56 57; 61 57 29; 62 58 59; 63 59 25; 64 60 61
41. 65 61 33; 66 62 63; 67 63 34; 68 64 65; 69 65 37; 70 66 67; 71 67 34; 72 68 69
42. 73 69 39; 74 70 71; 75 71 40; 76 72 73; 77 73 43; 78 74 75; 79 75 40; 80 52 68
43. 81 68 60; 82 60 44; 83 53 69; 84 69 61; 85 61 45; 86 54 70; 87 70 62; 88 62 46
44. 89 55 71; 90 71 63; 91 63 47; 92 56 72; 93 72 64; 94 64 48; 95 57 73; 96 73 65
45. 97 65 49; 98 58 74; 99 74 66; 100 66 50; 101 59 75; 102 75 67; 103 67 51
46. DEFINE PMEMBER
47. 3 48 49 5 50 51 PMEMBER 1
48. SUPPORTS
49. 14 20 23 27 30 31 35 41 PINNED
50. DEFINE MATERIAL START
51. ISOTROPIC ALUMINUM
52. E 6.89473E+07
53. POISSON 0.33
54. DENSITY 26.6017
55. ALPHA 1.28E-05
56. DAMP 0.03
57. ISOTROPIC STEEL
58. E 1.99947E+08
59. POISSON 0.3
60. DENSITY 76.8191
61. ALPHA 6.5E-06
62. DAMP 0.03
63. TYPE STEEL
64. STRENGTH RY 1.5 RT 1.2
65. END DEFINE MATERIAL
66. MEMBER PROPERTY TATASTRUCTURA
67. 1 3 5 TO 8 16 TO 21 25 27 29 32 34 35 37 39 41 44 46 TO 78 -
68. 79 TABLE ST 80X80X4.0SHS
69. 10 TO 12 14 15 24 26 28 31 33 36 38 40 43 45 80 TO 103 TABLE ST 25X25X2.0SHS
70. CONSTANTS
71. MATERIAL STEEL MEMB 1 3 5 TO 8 16 TO 21 25 27 29 32 34 35 37 39 41 44 46 TO 79
72. MATERIAL ALUMINUM MEMB 10 TO 12 14 15 24 26 28 31 33 36 38 40 43 45 80 TO 103
73. MEMBER RELEASE
74. 10 TO 12 14 15 24 26 28 31 33 36 38 40 43 45 80 TO 103 START MY MZ
75. 10 TO 12 14 15 24 26 28 31 33 36 38 40 43 45 80 TO 103 END MY MZ
76. DEFINE WIND LOAD

*** NOTE: If any floor diaphragm is present in the model Wind Load definition
          should be defined after Floor Diaphragm definition. Otherwise wind
          load generation may be unsuccessful during analysis.

77. TYPE 1 WIND 1
78. INT 1.27 1.4 1.45 1.59 1.74 1.95 HEIG  10  20  25  30  50  100
79. LOAD 1 LOADTYPE DEAD  TITLE DL
80. SELFWEIGHT Y -1
81. MEMBER LOAD
82. 25 27 29 32 34 35 37 39 41 44 46 47 64 TO 79 UNI GY -0.19
83. 1 3 5 TO 8 16 TO 21 48 TO 63 UNI GY -0.08
84. LOAD 2 LOADTYPE DEAD  TITLE LL
85. MEMBER LOAD
86. 27 29 34 35 39 41 46 47 64 TO 79 UNI GY -0.6
87. 3 5 7 8 17 18 20 21 48 TO 63 UNI GY -0.3
88. LOAD 3 LOADTYPE NONE  TITLE WL X
89. MEMBER LOAD
```

```
 90. 3 5 7 8 17 18 20 21 27 29 34 35 39 41 46 TO 79 UNI GY 0.9
 91. LOAD 4 LOADTYPE NONE  TITLE WL Z
 92. MEMBER LOAD
 93. 3 5 7 8 17 18 20 21 27 29 34 35 39 41 46 TO 79 UNI GY 0.9
 94. LOAD 5 LOADTYPE NONE  TITLE WL-X
 95. MEMBER LOAD
 96. 3 5 7 8 17 18 20 21 27 29 34 35 39 41 46 TO 79 UNI GY 0.9
 97. LOAD 6 LOADTYPE NONE  TITLE WL-Z
 98. MEMBER LOAD
 99. 3 5 7 8 17 18 20 21 27 29 34 35 39 41 46 TO 79 UNI GY 0.9
100. LOAD COMB 101 1.5(DL+LL)
101. 1 1.5 2 1.5
102. LOAD COMB 102 1.5(DL+WLX)
103. 1 1.5 3 1.5
104. LOAD COMB 103 1.5(DL+WLZ)
105. 1 1.5 4 1.5
106. LOAD COMB 104 1.5(DL+WL-X)
107. 1 1.5 5 1.5
108. LOAD COMB 105 1.5(DL+WL-Z)
109. 1 1.5 6 1.5
110. LOAD COMB 201 DL + LL
111. 1 1.0 2 1.0
112. LOAD COMB 202 DL+WLX
113. 1 1.0 3 1.0
114. LOAD COMB 203 DL+WLZ
115. 4 1.0 1 1.0
116. LOAD COMB 204 DL+WL-X
117. 1 1.0 5 1.0
118. LOAD COMB 205 DL+WL-Z
119. 6 1.0 1 1.0
120. LOAD COMB 206 DL+0.8WLX
121. 1 1.0 3 0.8
122. LOAD COMB 207 DL+0.8WLZ
123. 4 0.8 1 1.0
124. LOAD COMB 208 DL+0.8WL-X
125. 5 0.8 1 1.0
126. LOAD COMB 209 DL+0.8WL-Z
127. 6 0.8 1 1.0
128. PERFORM ANALYSIS PRINT ALL


         P R O B L E M   S T A T I S T I C S
         -----------------------------------

  NUMBER OF JOINTS         60  NUMBER OF MEMBERS      95
  NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS      8

        Using 64-bit analysis engine.

        SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER
```

STAAD SPACE -- PAGE NO. 4

ORIGINAL/FINAL BAND-WIDTH= 35/ 6/ 39 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 336
TOTAL LOAD COMBINATION CASES = 14 SO FAR.
SIZE OF STIFFNESS MATRIX = 14 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.3/ 204505.0 MB

STAAD SPACE -- PAGE NO. 5

LOADING 1 LOADTYPE DEAD TITLE DL

SELFWEIGHT Y -1.000

ACTUAL WEIGHT OF THE STRUCTURE = 4.216 KN

MEMBER LOAD - UNIT KN METE

MEMBER	UDL	L1	L2	CON	L	LIN1	LIN2
25	-0.1900 GY	0.00	0.67				
27	-0.1900 GY	0.00	0.87				
29	-0.1900 GY	0.00	0.75				
32	-0.1900 GY	0.00	0.67				
34	-0.1900 GY	0.00	0.87				
35	-0.1900 GY	0.00	0.75				
37	-0.1900 GY	0.00	0.67				
39	-0.1900 GY	0.00	0.87				
41	-0.1900 GY	0.00	0.75				
44	-0.1900 GY	0.00	0.67				
46	-0.1900 GY	0.00	0.87				
47	-0.1900 GY	0.00	0.75				
64	-0.1900 GY	0.00	0.87				
65	-0.1900 GY	0.00	0.87				
66	-0.1900 GY	0.00	0.75				
67	-0.1900 GY	0.00	0.75				
68	-0.1900 GY	0.00	0.87				
69	-0.1900 GY	0.00	0.87				
70	-0.1900 GY	0.00	0.75				
71	-0.1900 GY	0.00	0.75				
72	-0.1900 GY	0.00	0.87				
73	-0.1900 GY	0.00	0.87				
74	-0.1900 GY	0.00	0.75				
75	-0.1900 GY	0.00	0.75				
76	-0.1900 GY	0.00	0.87				
77	-0.1900 GY	0.00	0.87				
78	-0.1900 GY	0.00	0.75				
79	-0.1900 GY	0.00	0.75				
1	-0.0800 GY	0.00	0.67				
3	-0.0800 GY	0.00	0.87				
5	-0.0800 GY	0.00	0.75				
6	-0.0800 GY	0.00	0.67				
7	-0.0800 GY	0.00	0.87				
8	-0.0800 GY	0.00	0.75				
16	-0.0800 GY	0.00	0.67				
17	-0.0800 GY	0.00	0.87				
18	-0.0800 GY	0.00	0.75				
19	-0.0800 GY	0.00	0.67				
20	-0.0800 GY	0.00	0.87				
21	-0.0800 GY	0.00	0.75				
48	-0.0800 GY	0.00	0.87				
49	-0.0800 GY	0.00	0.87				

50	-0.0800	GY	0.00	0.75
51	-0.0800	GY	0.00	0.75
52	-0.0800	GY	0.00	0.87
53	-0.0800	GY	0.00	0.87
54	-0.0800	GY	0.00	0.75
55	-0.0800	GY	0.00	0.75
56	-0.0800	GY	0.00	0.87
57	-0.0800	GY	0.00	0.87
58	-0.0800	GY	0.00	0.75
59	-0.0800	GY	0.00	0.75
60	-0.0800	GY	0.00	0.87
61	-0.0800	GY	0.00	0.87
62	-0.0800	GY	0.00	0.75
63	-0.0800	GY	0.00	0.75

LOADING 2 LOADTYPE DEAD TITLE LL

MEMBER LOAD - UNIT KN METE

MEMBER	UDL		L1	L2	CON	L	LIN1	LIN2
27	-0.6000	GY	0.00	0.87				
29	-0.6000	GY	0.00	0.75				
34	-0.6000	GY	0.00	0.87				
35	-0.6000	GY	0.00	0.75				
39	-0.6000	GY	0.00	0.87				
41	-0.6000	GY	0.00	0.75				
46	-0.6000	GY	0.00	0.87				
47	-0.6000	GY	0.00	0.75				
64	-0.6000	GY	0.00	0.87				
65	-0.6000	GY	0.00	0.87				
66	-0.6000	GY	0.00	0.75				
67	-0.6000	GY	0.00	0.75				
68	-0.6000	GY	0.00	0.87				
69	-0.6000	GY	0.00	0.87				
70	-0.6000	GY	0.00	0.75				
71	-0.6000	GY	0.00	0.75				
72	-0.6000	GY	0.00	0.87				
73	-0.6000	GY	0.00	0.87				
74	-0.6000	GY	0.00	0.75				
75	-0.6000	GY	0.00	0.75				
76	-0.6000	GY	0.00	0.87				
77	-0.6000	GY	0.00	0.87				
78	-0.6000	GY	0.00	0.75				
79	-0.6000	GY	0.00	0.75				
3	-0.3000	GY	0.00	0.87				
5	-0.3000	GY	0.00	0.75				
7	-0.3000	GY	0.00	0.87				
8	-0.3000	GY	0.00	0.75				
17	-0.3000	GY	0.00	0.87				
18	-0.3000	GY	0.00	0.75				
20	-0.3000	GY	0.00	0.87				
21	-0.3000	GY	0.00	0.75				

48	-0.3000	GY	0.00	0.87
49	-0.3000	GY	0.00	0.87
50	-0.3000	GY	0.00	0.75
51	-0.3000	GY	0.00	0.75
52	-0.3000	GY	0.00	0.87
53	-0.3000	GY	0.00	0.87
54	-0.3000	GY	0.00	0.75
55	-0.3000	GY	0.00	0.75
56	-0.3000	GY	0.00	0.87
57	-0.3000	GY	0.00	0.87
58	-0.3000	GY	0.00	0.75
59	-0.3000	GY	0.00	0.75
60	-0.3000	GY	0.00	0.87
61	-0.3000	GY	0.00	0.87
62	-0.3000	GY	0.00	0.75
63	-0.3000	GY	0.00	0.75

LOADING 3 LOADTYPE NONE TITLE WL X

MEMBER LOAD - UNIT KN METE

MEMBER	UDL		L1	L2	CON	L	LIN1	LIN2
3	0.9000	GY	0.00	0.87				
5	0.9000	GY	0.00	0.75				
7	0.9000	GY	0.00	0.87				
8	0.9000	GY	0.00	0.75				
17	0.9000	GY	0.00	0.87				
18	0.9000	GY	0.00	0.75				
20	0.9000	GY	0.00	0.87				
21	0.9000	GY	0.00	0.75				
27	0.9000	GY	0.00	0.87				
29	0.9000	GY	0.00	0.75				
34	0.9000	GY	0.00	0.87				
35	0.9000	GY	0.00	0.75				
39	0.9000	GY	0.00	0.87				
41	0.9000	GY	0.00	0.75				
46	0.9000	GY	0.00	0.87				
47	0.9000	GY	0.00	0.75				
48	0.9000	GY	0.00	0.87				
49	0.9000	GY	0.00	0.87				
50	0.9000	GY	0.00	0.75				
51	0.9000	GY	0.00	0.75				
52	0.9000	GY	0.00	0.87				
53	0.9000	GY	0.00	0.87				
54	0.9000	GY	0.00	0.75				
55	0.9000	GY	0.00	0.75				
56	0.9000	GY	0.00	0.87				
57	0.9000	GY	0.00	0.87				
58	0.9000	GY	0.00	0.75				
59	0.9000	GY	0.00	0.75				
60	0.9000	GY	0.00	0.87				
61	0.9000	GY	0.00	0.87				

62	0.9000	GY	0.00	0.75
63	0.9000	GY	0.00	0.75
64	0.9000	GY	0.00	0.87
65	0.9000	GY	0.00	0.87
66	0.9000	GY	0.00	0.75
67	0.9000	GY	0.00	0.75
68	0.9000	GY	0.00	0.87
69	0.9000	GY	0.00	0.87
70	0.9000	GY	0.00	0.75
71	0.9000	GY	0.00	0.75
72	0.9000	GY	0.00	0.87
73	0.9000	GY	0.00	0.87
74	0.9000	GY	0.00	0.75
75	0.9000	GY	0.00	0.75
76	0.9000	GY	0.00	0.87
77	0.9000	GY	0.00	0.87
78	0.9000	GY	0.00	0.75
79	0.9000	GY	0.00	0.75

LOADING 4 LOADTYPE NONE TITLE WL Z

MEMBER LOAD - UNIT KN METE

MEMBER	UDL	L1	L2	CON	L	LIN1	LIN2
3	0.9000 GY	0.00	0.87				
5	0.9000 GY	0.00	0.75				
7	0.9000 GY	0.00	0.87				
8	0.9000 GY	0.00	0.75				
17	0.9000 GY	0.00	0.87				
18	0.9000 GY	0.00	0.75				
20	0.9000 GY	0.00	0.87				
21	0.9000 GY	0.00	0.75				
27	0.9000 GY	0.00	0.87				
29	0.9000 GY	0.00	0.75				
34	0.9000 GY	0.00	0.87				
35	0.9000 GY	0.00	0.75				
39	0.9000 GY	0.00	0.87				
41	0.9000 GY	0.00	0.75				
46	0.9000 GY	0.00	0.87				
47	0.9000 GY	0.00	0.75				
48	0.9000 GY	0.00	0.87				
49	0.9000 GY	0.00	0.87				
50	0.9000 GY	0.00	0.75				
51	0.9000 GY	0.00	0.75				
52	0.9000 GY	0.00	0.87				
53	0.9000 GY	0.00	0.87				
54	0.9000 GY	0.00	0.75				
55	0.9000 GY	0.00	0.75				
56	0.9000 GY	0.00	0.87				
57	0.9000 GY	0.00	0.87				
58	0.9000 GY	0.00	0.75				
59	0.9000 GY	0.00	0.75				

60	0.9000 GY	0.00	0.87
61	0.9000 GY	0.00	0.87
62	0.9000 GY	0.00	0.75
63	0.9000 GY	0.00	0.75
64	0.9000 GY	0.00	0.87
65	0.9000 GY	0.00	0.87
66	0.9000 GY	0.00	0.75
67	0.9000 GY	0.00	0.75
68	0.9000 GY	0.00	0.87
69	0.9000 GY	0.00	0.87
70	0.9000 GY	0.00	0.75
71	0.9000 GY	0.00	0.75
72	0.9000 GY	0.00	0.87
73	0.9000 GY	0.00	0.87
74	0.9000 GY	0.00	0.75
75	0.9000 GY	0.00	0.75
76	0.9000 GY	0.00	0.87
77	0.9000 GY	0.00	0.87
78	0.9000 GY	0.00	0.75
79	0.9000 GY	0.00	0.75

LOADING 5 LOADTYPE NONE TITLE WL-X

MEMBER LOAD - UNIT KN METE

MEMBER	UDL	L1	L2	CON	L	LIN1	LIN2
3	0.9000 GY	0.00	0.87				
5	0.9000 GY	0.00	0.75				
7	0.9000 GY	0.00	0.87				
8	0.9000 GY	0.00	0.75				
17	0.9000 GY	0.00	0.87				
18	0.9000 GY	0.00	0.75				
20	0.9000 GY	0.00	0.87				
21	0.9000 GY	0.00	0.75				
27	0.9000 GY	0.00	0.87				
29	0.9000 GY	0.00	0.75				
34	0.9000 GY	0.00	0.87				
35	0.9000 GY	0.00	0.75				
39	0.9000 GY	0.00	0.87				
41	0.9000 GY	0.00	0.75				
46	0.9000 GY	0.00	0.87				
47	0.9000 GY	0.00	0.75				
48	0.9000 GY	0.00	0.87				
49	0.9000 GY	0.00	0.87				
50	0.9000 GY	0.00	0.75				
51	0.9000 GY	0.00	0.75				
52	0.9000 GY	0.00	0.87				
53	0.9000 GY	0.00	0.87				
54	0.9000 GY	0.00	0.75				
55	0.9000 GY	0.00	0.75				
56	0.9000 GY	0.00	0.87				
57	0.9000 GY	0.00	0.87				

58	0.9000 GY	0.00	0.75
59	0.9000 GY	0.00	0.75
60	0.9000 GY	0.00	0.87
61	0.9000 GY	0.00	0.87
62	0.9000 GY	0.00	0.75
63	0.9000 GY	0.00	0.75
64	0.9000 GY	0.00	0.87
65	0.9000 GY	0.00	0.87
66	0.9000 GY	0.00	0.75
67	0.9000 GY	0.00	0.75
68	0.9000 GY	0.00	0.87
69	0.9000 GY	0.00	0.87
70	0.9000 GY	0.00	0.75
71	0.9000 GY	0.00	0.75
72	0.9000 GY	0.00	0.87
73	0.9000 GY	0.00	0.87
74	0.9000 GY	0.00	0.75
75	0.9000 GY	0.00	0.75
76	0.9000 GY	0.00	0.87
77	0.9000 GY	0.00	0.87
78	0.9000 GY	0.00	0.75
79	0.9000 GY	0.00	0.75

LOADING 6 LOADTYPE NONE TITLE WL-Z

MEMBER LOAD - UNIT KN METE

MEMBER	UDL	L1	L2	CON	L	LIN1	LIN2
3	0.9000 GY	0.00	0.87				
5	0.9000 GY	0.00	0.75				
7	0.9000 GY	0.00	0.87				
8	0.9000 GY	0.00	0.75				
17	0.9000 GY	0.00	0.87				
18	0.9000 GY	0.00	0.75				
20	0.9000 GY	0.00	0.87				
21	0.9000 GY	0.00	0.75				
27	0.9000 GY	0.00	0.87				
29	0.9000 GY	0.00	0.75				
34	0.9000 GY	0.00	0.87				
35	0.9000 GY	0.00	0.75				
39	0.9000 GY	0.00	0.87				
41	0.9000 GY	0.00	0.75				
46	0.9000 GY	0.00	0.87				
47	0.9000 GY	0.00	0.75				
48	0.9000 GY	0.00	0.87				
49	0.9000 GY	0.00	0.87				
50	0.9000 GY	0.00	0.75				
51	0.9000 GY	0.00	0.75				
52	0.9000 GY	0.00	0.87				
53	0.9000 GY	0.00	0.87				
54	0.9000 GY	0.00	0.75				
55	0.9000 GY	0.00	0.75				

56	0.9000 GY	0.00	0.87
57	0.9000 GY	0.00	0.87
58	0.9000 GY	0.00	0.75
59	0.9000 GY	0.00	0.75
60	0.9000 GY	0.00	0.87
61	0.9000 GY	0.00	0.87
62	0.9000 GY	0.00	0.75
63	0.9000 GY	0.00	0.75
64	0.9000 GY	0.00	0.87
65	0.9000 GY	0.00	0.87
66	0.9000 GY	0.00	0.75
67	0.9000 GY	0.00	0.75
68	0.9000 GY	0.00	0.87
69	0.9000 GY	0.00	0.87
70	0.9000 GY	0.00	0.75
71	0.9000 GY	0.00	0.75
72	0.9000 GY	0.00	0.87
73	0.9000 GY	0.00	0.87
74	0.9000 GY	0.00	0.75
75	0.9000 GY	0.00	0.75
76	0.9000 GY	0.00	0.87
77	0.9000 GY	0.00	0.87
78	0.9000 GY	0.00	0.75
79	0.9000 GY	0.00	0.75

***WARNING - INSTABILITY AT JOINT 27 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.0853067E+04 L-MATRIX DIAG= 2.4556357E-10 EQN NO 334
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

FOR LOADING - 1
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
14	0.00000E+00	-5.66123E-02	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
15	-3.71642E-10	-1.33411E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.05619E-02
18	4.87279E-10	-1.30784E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.07083E-07
19	-5.35942E-10	-1.40722E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.57900E-03
20	0.00000E+00	-5.66123E-02	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
21	-5.16100E-10	-1.33411E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.05619E-02
22	3.42311E-10	-1.40722E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.57899E-03
23	0.00000E+00	-5.66123E-02	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
24	-3.71642E-10	-1.33411E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.05619E-02
25	4.87279E-10	-1.30784E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.07083E-07
26	-5.35942E-10	-1.40722E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.57900E-03
27	0.00000E+00	-5.66123E-02	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00

STAAD SPACE -- PAGE NO. 12

APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
28	-5.16100E-10	-1.33411E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.05619E-02
29	3.42311E-10	-1.40722E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.57899E-03
30	0.00000E+00	-9.31873E-02	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
31	0.00000E+00	-9.31873E-02	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
32	1.71918E-09	-2.20643E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.73856E-02
33	-1.26353E-10	-2.32677E-01	0.00000E+00	0.00000E+00	0.00000E+00	4.24519E-03
34	5.83094E-10	-2.16319E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.38405E-07
35	0.00000E+00	-9.31873E-02	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
36	-4.51932E-10	-2.20643E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.73855E-02
37	-1.44341E-09	-2.32677E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.24518E-03
38	1.71918E-09	-2.20643E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.73856E-02
39	-1.26353E-10	-2.32677E-01	0.00000E+00	0.00000E+00	0.00000E+00	4.24519E-03
40	5.83094E-10	-2.16319E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.38405E-07
41	0.00000E+00	-9.31873E-02	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
42	-4.51932E-10	-2.20643E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.73855E-02
43	-1.44341E-09	-2.32677E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.24518E-03
44	-1.27245E-09	-1.50658E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.12802E-07
45	-1.32603E-09	-1.50659E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.04733E-07
46	6.30342E-11	-1.30783E-01	0.00000E+00	0.00000E+00	0.00000E+00	8.41803E-09
47	5.50313E-10	-1.30784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.15501E-07
48	1.48784E-09	-1.50658E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.68361E-09
49	2.23562E-09	-1.50659E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.44965E-07
50	-3.07767E-10	-1.30784E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.07083E-07
51	2.21878E-10	-1.30784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.07083E-07
52	-1.27245E-09	-1.50658E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.12802E-07
53	-1.32603E-09	-1.50659E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.04733E-07
54	6.30342E-11	-1.30783E-01	0.00000E+00	0.00000E+00	0.00000E+00	8.41803E-09
55	5.50313E-10	-1.30784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.15501E-07
56	1.48784E-09	-1.50658E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.68361E-09
57	2.23562E-09	-1.50659E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.44965E-07
58	-3.07767E-10	-1.30784E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.07083E-07
59	2.21878E-10	-1.30784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-2.07083E-07
60	-1.04542E-09	-2.49032E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.86880E-07
61	-6.42684E-10	-2.49033E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.05082E-07
62	8.36708E-10	-2.16318E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.43106E-08
63	1.41980E-09	-2.16319E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.53557E-07
64	7.04376E-10	-2.49032E-01	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
65	2.37976E-09	-2.49034E-01	0.00000E+00	0.00000E+00	0.00000E+00	4.04065E-07
66	-6.09067E-10	-2.16319E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.39246E-07
67	-6.09067E-10	-2.16319E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.39246E-07
68	-1.04542E-09	-2.49032E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.86880E-07
69	-6.42684E-10	-2.49033E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.05082E-07
70	8.36708E-10	-2.16318E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.43106E-08
71	1.41980E-09	-2.16319E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.53557E-07
72	7.04376E-10	-2.49032E-01	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
73	2.37976E-09	-2.49034E-01	0.00000E+00	0.00000E+00	0.00000E+00	4.04065E-07
74	-6.09067E-10	-2.16319E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.39246E-07
75	-6.09067E-10	-2.16319E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.39246E-07

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE DEAD TITLE DL

STAAD SPACE -- PAGE NO. 13

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.482200986E+01
Y = 0.790425168E+01
Z = -0.295500496E+01

TOTAL APPLIED LOAD 1

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -10.18
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -30.07 MY= -0.00 MZ= -49.08

TOTAL REACTION LOAD 1

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 10.18
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 30.07 MY= 0.00 MZ= 49.08

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	1.78523E-01	36
Y =	-1.59287E+00	40
Z =	0.00000E+00	0
RX=	0.00000E+00	0
RY=	0.00000E+00	0
RZ=	5.11928E-03	65

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
14	0.00	-0.06	0.00	0.00	0.00	0.00	
	-1.41	-0.90	0.00	0.00	0.00	-0.00	111000
20	0.00	-0.06	0.00	0.00	0.00	0.00	
	1.41	-0.90	0.00	0.00	0.00	0.00	111000
23	0.00	-0.06	0.00	0.00	0.00	0.00	
	-1.41	-0.90	0.00	0.00	0.00	0.00	111000
27	0.00	-0.06	0.00	0.00	0.00	0.00	
	1.41	-0.90	0.00	0.00	0.00	-0.00	111000
30	0.00	-0.09	0.00	0.00	0.00	0.00	
	-2.33	-1.49	0.00	0.00	0.00	-0.00	111000
31	0.00	-0.09	0.00	0.00	0.00	0.00	
	-2.33	-1.49	0.00	0.00	0.00	-0.00	111000
35	0.00	-0.09	0.00	0.00	0.00	0.00	
	2.33	-1.49	0.00	0.00	0.00	-0.00	111000

STAAD SPACE -- PAGE NO. 14

41	0.00	-0.09	0.00	0.00	0.00	0.00	
	2.33	-1.49	0.00	0.00	0.00	0.00	111000

FOR LOADING - 2
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
15	5.99289E-10	-1.30139E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.86100E-02
18	3.09465E-10	-2.25261E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.63659E-07
19	1.14224E-09	-2.42771E-01	0.00000E+00	0.00000E+00	0.00000E+00	4.54416E-03
21	2.03777E-09	-1.30139E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.86099E-02
22	-1.76684E-09	-2.42771E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.54415E-03
24	5.99289E-10	-1.30139E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.86100E-02
25	3.09465E-10	-2.25261E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.63659E-07
26	1.14224E-09	-2.42771E-01	0.00000E+00	0.00000E+00	0.00000E+00	4.54416E-03
28	2.03777E-09	-1.30139E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.86099E-02
29	-1.76684E-09	-2.42771E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.54415E-03
32	1.19858E-09	-2.60279E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.72200E-02
33	2.28447E-09	-4.85541E-01	0.00000E+00	0.00000E+00	0.00000E+00	9.08831E-03
34	6.18930E-10	-4.50523E-01	0.00000E+00	0.00000E+00	0.00000E+00	7.27317E-07
36	4.07553E-09	-2.60278E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.72198E-02
37	-3.53368E-09	-4.85541E-01	0.00000E+00	0.00000E+00	0.00000E+00	-9.08830E-03
38	1.19858E-09	-2.60279E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.72200E-02
39	2.28447E-09	-4.85541E-01	0.00000E+00	0.00000E+00	0.00000E+00	9.08831E-03
40	6.18930E-10	-4.50523E-01	0.00000E+00	0.00000E+00	0.00000E+00	7.27317E-07
42	4.07553E-09	-2.60278E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.72198E-02
43	-3.53368E-09	-4.85541E-01	0.00000E+00	0.00000E+00	0.00000E+00	-9.08830E-03
44	-1.72920E-09	-2.60278E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.98665E-07
45	1.71397E-09	-2.60279E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.40437E-07
46	-1.16437E-09	-2.25260E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.43106E-08
47	2.04397E-10	-2.25261E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.79653E-07
48	-1.35292E-09	-2.60278E-01	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
49	-3.14261E-09	-2.60280E-01	0.00000E+00	0.00000E+00	0.00000E+00	4.31003E-07
50	3.34067E-10	-2.25261E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.63659E-07
51	3.34067E-10	-2.25261E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.63659E-07
52	-1.72920E-09	-2.60278E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.98665E-07
53	1.71397E-09	-2.60279E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.40437E-07
54	-1.16437E-09	-2.25260E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.43106E-08
55	2.04397E-10	-2.25261E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.79653E-07
56	-1.35292E-09	-2.60278E-01	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
57	-3.14261E-09	-2.60280E-01	0.00000E+00	0.00000E+00	0.00000E+00	4.31003E-07
58	3.34067E-10	-2.25261E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.63659E-07
59	3.34067E-10	-2.25261E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.63659E-07
60	-3.45839E-09	-5.20556E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.97331E-07
61	3.42795E-09	-5.20559E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08087E-06
62	-2.32874E-09	-4.50519E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.86213E-08
63	4.08794E-10	-4.50522E-01	0.00000E+00	0.00000E+00	0.00000E+00	-7.59306E-07
64	-2.70584E-09	-5.20556E-01	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
65	-6.28523E-09	-5.20559E-01	0.00000E+00	0.00000E+00	0.00000E+00	8.62006E-07
66	6.68134E-10	-4.50523E-01	0.00000E+00	0.00000E+00	0.00000E+00	7.27317E-07
67	6.68134E-10	-4.50523E-01	0.00000E+00	0.00000E+00	0.00000E+00	-7.27317E-07
68	-3.45839E-09	-5.20556E-01	0.00000E+00	0.00000E+00	0.00000E+00	3.97331E-07
69	3.42795E-09	-5.20559E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08087E-06
70	-2.32874E-09	-4.50519E-01	0.00000E+00	0.00000E+00	0.00000E+00	2.86213E-08
71	4.08794E-10	-4.50522E-01	0.00000E+00	0.00000E+00	0.00000E+00	-7.59306E-07
72	-2.70584E-09	-5.20556E-01	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
73	-6.28523E-09	-5.20559E-01	0.00000E+00	0.00000E+00	0.00000E+00	8.62006E-07

STAAD SPACE -- PAGE NO. 15

APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
74	6.68134E-10	-4.50523E-01	0.00000E+00	0.00000E+00	0.00000E+00	7.27317E-07
75	6.68134E-10	-4.50523E-01	0.00000E+00	0.00000E+00	0.00000E+00	-7.27317E-07

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE DEAD TITLE LL

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.482200990E+01
Y = 0.798095396E+01
Z = -0.295500496E+01

TOTAL APPLIED LOAD 2

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -17.48
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -51.65 MY= 0.00 MZ= -84.29

TOTAL REACTION LOAD 2

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 17.48
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 51.65 MY= -0.00 MZ= 84.29

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	3.71907E-01	42
Y =	-3.31780E+00	40
Z =	0.00000E+00	0
RX=	0.00000E+00	0
RY=	0.00000E+00	0
RZ=	1.06647E-02	73

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
14	0.00	0.00	0.00	0.00	0.00	0.00	
	-2.44	-1.46	0.00	0.00	0.00	-0.00	111000
20	0.00	0.00	0.00	0.00	0.00	0.00	
	2.44	-1.46	0.00	0.00	0.00	-0.00	111000
23	0.00	0.00	0.00	0.00	0.00	0.00	
	-2.44	-1.46	0.00	0.00	0.00	0.00	111000

STAAD SPACE -- PAGE NO. 16

27	0.00	0.00	0.00	0.00	0.00	0.00	
	2.44	-1.46	0.00	0.00	0.00	0.00	111000
30	0.00	0.00	0.00	0.00	0.00	0.00	
	-4.86	-2.91	0.00	0.00	0.00	0.00	111000
31	0.00	0.00	0.00	0.00	0.00	0.00	
	-4.86	-2.91	0.00	0.00	0.00	0.00	111000
35	0.00	0.00	0.00	0.00	0.00	0.00	
	4.86	-2.91	0.00	0.00	0.00	-0.00	111000
41	0.00	0.00	0.00	0.00	0.00	0.00	
	4.86	-2.91	0.00	0.00	0.00	0.00	111000

FOR LOADING - 3
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
15	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
18	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
19	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
21	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
22	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
24	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
25	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
26	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
28	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
29	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
32	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
33	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
34	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
36	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
37	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
38	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
39	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
40	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
42	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
43	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
44	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
45	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
46	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
47	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
48	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
49	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
50	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
51	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06
52	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
53	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
54	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
55	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
56	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
57	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
58	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
59	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06
60	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
61	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
62	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
63	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
64	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
65	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06

STAAD SPACE -- PAGE NO. 17

APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
66	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
67	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06
68	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
69	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
70	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
71	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
72	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
73	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
74	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
75	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE NONE TITLE WL X

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.482200988E+01
Y = 0.798095396E+01
Z = -0.295500496E+01

TOTAL APPLIED LOAD 3

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 34.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 103.30 MY= 0.00 MZ= 168.57

TOTAL REACTION LOAD 3

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -34.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -103.30 MY= -0.00 MZ= -168.57

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	-5.58940E-01	21
Y =	4.98705E+00	18
Z =	0.00000E+00	0
RX=	0.00000E+00	0
RY=	0.00000E+00	0
RZ=	-1.60292E-02	49

STAAD SPACE -- PAGE NO. 18

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
14	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	0.00	111000
20	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	-0.00	111000
23	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	-0.00	111000
27	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	0.00	111000
30	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	0.00	111000
31	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	-0.00	111000
35	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	0.00	111000
41	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	0.00	111000

FOR LOADING - 4
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
15	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
18	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
19	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
21	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
22	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
24	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
25	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
26	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
28	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
29	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
32	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
33	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
34	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
36	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
37	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
38	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
39	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
40	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
42	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
43	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
44	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
45	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
46	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
47	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
48	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
49	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
50	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
51	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06
52	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07

STAAD SPACE -- PAGE NO. 19

APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
53	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
54	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
55	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
56	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
57	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
58	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
59	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06
60	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
61	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
62	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
63	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
64	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
65	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
66	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
67	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06
68	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
69	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
70	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
71	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
72	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
73	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
74	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
75	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
LOADTYPE NONE TITLE WL Z

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.482200988E+01
Y = 0.798095396E+01
Z = -0.295500496E+01

TOTAL APPLIED LOAD 4

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 34.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 103.30 MY= 0.00 MZ= 168.57

TOTAL REACTION LOAD 4

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -34.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -103.30 MY= -0.00 MZ= -168.57

STAAD SPACE -- PAGE NO. 20

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	-5.58940E-01	21
Y =	4.98705E+00	18
Z =	0.00000E+00	0
RX=	0.00000E+00	0
RY=	0.00000E+00	0
RZ=	-1.60292E-02	49

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
14	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	0.00	111000
20	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	-0.00	111000
23	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	-0.00	111000
27	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	0.00	111000
30	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	0.00	111000
31	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	-0.00	111000
35	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	0.00	111000
41	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	0.00	111000

FOR LOADING - 5
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
15	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
18	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
19	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
21	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
22	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
24	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
25	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
26	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
28	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
29	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
32	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
33	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
34	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
36	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
37	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
38	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
39	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
40	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
42	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
43	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02

STAAD SPACE -- PAGE NO. 21

APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
44	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
45	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
46	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
47	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
48	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
49	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
50	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
51	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06
52	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
53	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
54	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
55	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
56	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
57	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
58	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
59	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06
60	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
61	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
62	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
63	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
64	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
65	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
66	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
67	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06
68	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
69	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
70	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
71	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
72	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
73	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
74	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
75	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
LOADTYPE NONE TITLE WL-X

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.482200988E+01
Y = 0.798095396E+01
Z = -0.295500496E+01

TOTAL APPLIED LOAD 5

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 34.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 103.30 MY= 0.00 MZ= 168.57

STAAD SPACE -- PAGE NO. 22

TOTAL REACTION LOAD 5

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -34.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -103.30 MY= -0.00 MZ= -168.57

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)
MAXIMUMS AT NODE
X = -5.58940E-01 21
Y = 4.98705E+00 18
Z = 0.00000E+00 0
RX= 0.00000E+00 0
RY= 0.00000E+00 0
RZ= -1.60292E-02 49

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
14	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	0.00	111000
20	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	-0.00	111000
23	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	-0.00	111000
27	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	0.00	111000
30	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	0.00	111000
31	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	-0.00	111000
35	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	0.00	111000
41	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	0.00	111000

FOR LOADING - 6
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
15	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
18	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
19	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
21	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
22	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
24	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
25	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
26	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
28	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
29	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
32	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02

STAAD SPACE -- PAGE NO. 23

APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
33	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
34	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
36	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
37	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
38	-2.20990E-09	3.90418E-01	0.00000E+00	0.00000E+00	0.00000E+00	5.58299E-02
39	5.31511E-09	7.28312E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.36325E-02
40	1.24170E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
42	-2.42549E-09	3.90417E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.58296E-02
43	2.70261E-09	7.28311E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.36325E-02
44	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
45	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
46	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
47	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
48	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
49	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
50	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
51	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06
52	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
53	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
54	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
55	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
56	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
57	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
58	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
59	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06
60	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
61	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
62	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
63	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
64	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
65	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
66	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
67	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06
68	-1.41666E-10	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-5.99363E-07
69	-8.01214E-09	7.80838E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.62636E-06
70	9.09203E-10	6.75779E-01	0.00000E+00	0.00000E+00	0.00000E+00	-4.71409E-08
71	-2.08632E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.13812E-06
72	3.24401E-09	7.80834E-01	0.00000E+00	0.00000E+00	0.00000E+00	-3.36721E-09
73	4.10713E-09	7.80839E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.29638E-06
74	-5.36867E-10	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	-1.08761E-06
75	-4.77414E-09	6.75784E-01	0.00000E+00	0.00000E+00	0.00000E+00	1.08761E-06

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
LOADTYPE NONE TITLE WL-Z

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.482200988E+01
Y = 0.798095396E+01
Z = -0.295500496E+01

STAAD SPACE -- PAGE NO. 24

TOTAL APPLIED LOAD 6

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 34.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 103.30 MY= 0.00 MZ= 168.57

TOTAL REACTION LOAD 6

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -34.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -103.30 MY= -0.00 MZ= -168.57

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X =	-5.58940E-01	21
Y =	4.98705E+00	18
Z =	0.00000E+00	0
RX=	0.00000E+00	0
RY=	0.00000E+00	0
RZ=	-1.60292E-02	49

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
14	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	0.00	111000
20	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	-0.00	111000
23	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	-0.00	111000
27	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	0.00	111000
30	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	0.00	111000
31	0.00	0.00	0.00	0.00	0.00	0.00	
	7.30	4.37	0.00	0.00	0.00	-0.00	111000
35	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	0.00	111000
41	0.00	0.00	0.00	0.00	0.00	0.00	
	-7.30	4.37	0.00	0.00	0.00	0.00	111000

LOAD COMBINATION NO. 101
1.5(DL+LL)

LOAD COMBINATION NO. 102
1.5(DL+WLX)

LOADING-	1.	3.
FACTOR -	1.50	1.50

LOAD COMBINATION NO. 103
1.5(DL+WLZ)

LOADING-	1.	4.
FACTOR -	1.50	1.50

LOAD COMBINATION NO. 104
1.5(DL+WL-X)

LOADING-	1.	5.
FACTOR -	1.50	1.50

LOAD COMBINATION NO. 105
1.5(DL+WL-Z)

LOADING-	1.	6.
FACTOR -	1.50	1.50

LOAD COMBINATION NO. 201
DL + LL

LOADING-	1.	2.
FACTOR -	1.00	1.00

LOAD COMBINATION NO. 202
DL+WLX

LOADING-	1.	3.
FACTOR -	1.00	1.00

LOAD COMBINATION NO. 203
DL+WLZ

LOADING-	4.	1.
FACTOR -	1.00	1.00

LOAD COMBINATION NO. 204
DL+WL-X

LOADING-	1.	5.
FACTOR -	1.00	1.00

LOAD COMBINATION NO. 205
DL+WL-Z

LOADING-	6.	1.
FACTOR -	1.00	1.00

LOAD COMBINATION NO. 206
DL+0.8WLX

LOADING- 1. 3.
FACTOR - 1.00 0.80

LOAD COMBINATION NO. 207
DL+0.8WLZ

LOADING- 4. 1.
FACTOR - 0.80 1.00

LOAD COMBINATION NO. 208
DL+0.8WL-X

LOADING- 5. 1.
FACTOR - 0.80 1.00

LOAD COMBINATION NO. 209
DL+0.8WL-Z

LOADING- 6. 1.
FACTOR - 0.80 1.00

********** END OF DATA FROM INTERNAL STORAGE **********

129. PARAMETER 1
130. CODE IS800 LSD
131. FYLD 310000 MEMB 1 3 5 TO 8 16 TO 21 25 27 29 32 34 35 37 39 41 44 46 TO 79
132. CHECK CODE MEMB 1 3 5 TO 8 16 TO 21 25 27 29 32 34 35 37 39 41 44 46 TO 79
STEEL DESIGN

Member Number: 1					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS Ratio: 0.630 Critical Load Case: 102 Location:					0.67
Critical Condition: Sec. 8.2.1.2					

Member Number: 3					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS Ratio: 0.630 Critical Load Case: 102 Location:					0.00
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 27

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 5					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.192	Critical Load Case:	102	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

Member Number: 6					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.630	Critical Load Case:	102	Location:	0.67
Critical Condition: Sec. 8.2.1.2					

Member Number: 7					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.630	Critical Load Case:	102	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 8					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.192	Critical Load Case:	102	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 28

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 16					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.630	Critical Load Case:	102	Location:	0.67
Critical Condition: Sec. 8.2.1.2					

Member Number: 17					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.630	Critical Load Case:	102	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 18					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.192	Critical Load Case:	102	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

Member Number: 19					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.630	Critical Load Case:	102	Location:	0.67
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 29

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 20					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.630	Critical Load Case:	102	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 21					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.192	Critical Load Case:	102	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

Member Number: 25					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.769	Critical Load Case:	101	Location:	0.67
Critical Condition: Sec. 8.2.1.2					

Member Number: 27					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.769	Critical Load Case:	101	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 30

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 29					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.235	Critical Load Case:	101	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

Member Number: 32					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.769	Critical Load Case:	101	Location:	0.67
Critical Condition: Sec. 8.2.1.2					

Member Number: 34					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.769	Critical Load Case:	101	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 35					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.235	Critical Load Case:	101	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 37					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.769	Critical Load Case: 101	Location:	0.67	
Critical Condition: Sec. 8.2.1.2					

Member Number: 39					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.769	Critical Load Case: 101	Location:	0.00	
Critical Condition: Sec. 8.2.1.2					

Member Number: 41					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.235	Critical Load Case: 101	Location:	0.75	
Critical Condition: Sec. 8.2.1.2					

Member Number: 44					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.769	Critical Load Case: 101	Location:	0.67	
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 32

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 46					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.769	Critical Load Case:	101	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 47					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.235	Critical Load Case:	101	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

Member Number: 48					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.308	Critical Load Case:	102	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 49					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.074	Critical Load Case:	102	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 33

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 50					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.245	Critical Load Case:	102	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

Member Number: 51					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.248	Critical Load Case:	102	Location:	0.25
Critical Condition: Sec. 8.2.1.2					

Member Number: 52					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.308	Critical Load Case:	102	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 53					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.074	Critical Load Case:	102	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 34

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 54					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.245	Critical Load Case:	102	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

Member Number: 55					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.248	Critical Load Case:	102	Location:	0.25
Critical Condition: Sec. 8.2.1.2					

Member Number: 56					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.308	Critical Load Case:	102	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 57					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.074	Critical Load Case:	102	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 35

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 58					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.245	Critical Load Case:	102	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

Member Number: 59					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.248	Critical Load Case:	102	Location:	0.25
Critical Condition: Sec. 8.2.1.2					

Member Number: 60					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.308	Critical Load Case:	102	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 61					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.074	Critical Load Case:	102	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 62					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.245	Critical Load Case:	102	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

Member Number: 63					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.248	Critical Load Case:	102	Location:	0.25
Critical Condition: Sec. 8.2.1.2					

Member Number: 64					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.377	Critical Load Case:	101	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 65					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.090	Critical Load Case:	101	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 37

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 66					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.300	Critical Load Case:	101	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

Member Number: 67					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.303	Critical Load Case:	101	Location:	0.25
Critical Condition: Sec. 8.2.1.2					

Member Number: 68					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.377	Critical Load Case:	101	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 69					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.090	Critical Load Case:	101	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 38

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 70					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.300	Critical Load Case:	101	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

Member Number: 71					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.303	Critical Load Case:	101	Location:	0.25
Critical Condition: Sec. 8.2.1.2					

Member Number: 72					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.377	Critical Load Case:	101	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 73					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.090	Critical Load Case:	101	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 39

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 74					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.300	Critical Load Case:	101	Location:	0.75
Critical Condition: Sec. 8.2.1.2					

Member Number: 75					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.303	Critical Load Case:	101	Location:	0.25
Critical Condition: Sec. 8.2.1.2					

Member Number: 76					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.377	Critical Load Case:	101	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 77					
Member Section: ST 80X80X4.0SHS (TATA STRUCTURA SECTIONS)					
Status: PASS	Ratio: 0.090	Critical Load Case:	101	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 40

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.1)
****

Member Number: 78						
Member Section: ST	80X80X4.0SHS	(TATA STRUCTURA SECTIONS)				
Status: PASS	Ratio: 0.300	Critical Load Case:	101	Location:	0.75	
Critical Condition: Sec. 8.2.1.2						

Member Number: 79						
Member Section: ST	80X80X4.0SHS	(TATA STRUCTURA SECTIONS)				
Status: PASS	Ratio: 0.303	Critical Load Case:	101	Location:	0.25	
Critical Condition: Sec. 8.2.1.2						

133. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= AUG 10,2023 TIME= 11:31:22 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://www.bentley.com/en/support/                      *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2017 Bentley Systems, Inc.        *
*                http://www.bentley.com                     *
*************************************************************
```